**Subject:** Submission to Journal of Mathematical Biology

Dear Editors,

We submit the manuscript entitled *A PDE-Constrained Hodge–Laplacian Framework for Falsifiable Inference in Transport-Dominated Systems* for consideration in the *Journal of Mathematical Biology*.

This work introduces a mathematically grounded framework for falsifiable inference of transport processes on discrete domains. The central contribution is an operator-level formulation in which advection–diffusion dynamics and conservation laws are enforced through discrete exterior calculus and Hodge–Laplacian operators, yielding explicit integral constraints derived from Green–Stokes relations. Model adequacy is assessed via conservation residuals rather than predictive error, providing a falsifiable criterion for mechanistic compatibility.

Methodologically, the framework bridges finite element exterior calculus, PDE-constrained optimization, and graph-based representations. Importantly, the approach is not intended as a universal biological model; instead, advection–diffusion is formalized as a mathematically precise null model of passive transport. Structured deviations from this null model are diagnosed through typed residuals obtained via Hodge decomposition, enabling discrimination between divergence-dominated, curl-dominated, and harmonic failure modes.

While motivated by biological transport, the manuscript emphasizes mathematical structure, identifiability, and robustness. All validation is conducted on controlled synthetic systems, supplemented by a minimal real-world diagnostic example included solely to demonstrate applicability under irregular sampling, without biological interpretation or claims.

We believe this contribution aligns with the aims of the *Journal of Mathematical Biology* by advancing mathematically principled tools for mechanistic modeling, falsifiability, and diagnostic analysis of transport phenomena on discrete geometries.

This manuscript has not been published elsewhere and is not under consideration by another journal.

Sincerely,

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